CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

Chapter 3: Pair of Linear Equations in Two Variables

Topic-wise short notes with solved examples

Two linear equations in the same two variables are two straight lines. Solving the pair means finding the point, if any, that lies on both lines.

1 The pair and its solution

- A **linear equation in two variables** has the form $ax + by + c = 0$, where a and b are not both zero. Its graph is a straight line.
- A **pair of linear equations** is written as

$$a_1x + b_1y + c_1 = 0 \qquad a_2x + b_2y + c_2 = 0$$

- A **solution** of the pair is a pair of values (x, y) that satisfies both equations.
- A pair that has at least one solution is called **consistent**. A pair with no solution is **inconsistent**.

2 Graphical method

- Draw both lines on the same axes. Find at least two points on each line; a third point is a useful check.
- **Intersecting lines**: exactly one solution, the point of intersection.
- **Parallel lines**: no solution.
- **Coincident lines** (one line lying on the other): infinitely many solutions.

Example 2.1 Solve graphically: $x + y = 5$ and $x - y = 1$.

SOLUTION

Points on $x + y = 5$: $(0, 5)$, $(5, 0)$, $(2, 3)$

Points on $x - y = 1$: $(1, 0)$, $(0, -1)$, $(4, 3)$

Plot both lines. They intersect at the point $(3, 2)$.

Check: $3 + 2 = 5$ and $3 - 2 = 1$.

$x = 3, y = 2$

3 Conditions for consistency

- Compare the ratios of the coefficients. No graph is needed.

Ratios	Lines	Solutions	The pair is
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	Intersecting	Exactly one	Consistent
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	Parallel	None	Inconsistent
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	Coincident	Infinitely many	Consistent (dependent)

Example 3.1 Without drawing graphs, say whether each pair is consistent: (i) $2x + 3y = 7$, $4x + 6y = 14$ (ii) $x + 2y = 4$, $2x + 4y = 12$.

SOLUTION

(i) $\frac{a_1}{a_2} = \frac{2}{4} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{3}{6} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{7}{14} = \frac{1}{2}$. All three are equal, so the lines coincide.

(ii) $\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{2}{4} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{4}{12} = \frac{1}{3}$. The first two are equal but not the third, so the lines are parallel.

(i) Consistent, with infinitely many solutions. (ii) Inconsistent, with no solution.

Example 3.2 For which value of k does the pair $2x + 3y = 7$, $(k-1)x + (k+2)y = 3k$ have infinitely many solutions?

SOLUTION

We need $\frac{2}{k-1} = \frac{3}{k+2} = \frac{7}{3k}$.

From the first two: $2(k+2) = 3(k-1)$, so $2k + 4 = 3k - 3$ and $k = 7$.

Check with the third: $\frac{3}{9} = \frac{1}{3}$ and $\frac{7}{21} = \frac{1}{3}$. It holds.

$k = 7$

REMEMBER

Write both equations in the same form, with the terms in the same order, before reading off a, b and c.

4 Substitution method

- Step 1: from one equation, express one variable in terms of the other.
- Step 2: substitute this in the other equation to get an equation in one variable, and solve it.
- Step 3: put the value back to find the other variable.

Example 4.1 Solve by substitution: $2x + y = 11$ and $x - 3y = -5$.

SOLUTION

From the first equation: $y = 11 - 2x$

Substitute in the second: $x - 3(11 - 2x) = -5$, so $7x - 33 = -5$ and $x = 4$

Then $y = 11 - 2(4) = 3$

$x = 4, y = 3$

5 Elimination method

- Step 1: multiply the equations by suitable numbers so that the coefficients of one variable become equal in size.
- Step 2: add or subtract the equations to remove that variable, and solve.
- Step 3: substitute to find the other variable.

Example 5.1 Solve by elimination: $3x + 2y = 16$ and $5x - 3y = 14$.

SOLUTION

Multiply the first equation by 3 and the second by 2: $9x + 6y = 48$ and $10x - 6y = 28$

Add: $19x = 76$, so $x = 4$

Put $x = 4$ in $3x + 2y = 16$: $12 + 2y = 16$, so $y = 2$

$x = 4, y = 2$

REMEMBER

If the working ends in a true statement such as $0 = 0$, the pair has infinitely many solutions. If it ends in a false statement such as $0 = 5$, the pair has no solution.

6 Word problems

- Choose the two unknowns as x and y, write one equation for each condition, solve, and check the answer against the words of the problem.
- A two-digit number with tens digit x and units digit y is $10x + y$. With the digits reversed it is $10y + x$.
- A fixed charge with a charge per unit: total $= x + (\text{number of units}) \times y$.

Example 6.1 2 notebooks and 3 pens cost ₹130. 3 notebooks and 2 pens cost ₹170. Find the cost of one notebook and of one pen.

SOLUTION

Let a notebook cost ₹x and a pen cost ₹y. Then $2x + 3y = 130$ and $3x + 2y = 170$.

Add: $5x + 5y = 300$, so $x + y = 60$. Subtract: $x - y = 40$.

Adding these two: $2x = 100$, so $x = 50$ and $y = 10$.

A notebook costs ₹50 and a pen costs ₹10.

Example 6.2 The sum of the digits of a two-digit number is 9. The number obtained by reversing the digits is 27 more than the number. Find the number.

SOLUTION

Let the tens digit be x and the units digit be y. Then $x + y = 9$.

$(10y + x) - (10x + y) = 27$, so $9y - 9x = 27$ and $y - x = 3$.

Solving: $y = 6$, $x = 3$.

The number is 36.

Example 6.3 A taxi charges a fixed amount plus a charge per km. A 12 km journey costs ₹210 and a 20 km journey costs ₹330. Find the fixed charge and the charge per km.

SOLUTION

Let the fixed charge be ₹x and the charge per km be ₹y. Then $x + 12y = 210$ and $x + 20y = 330$.

Subtract: $8y = 120$, so $y = 15$. Then $x = 210 - 180 = 30$.

Fixed charge ₹30; ₹15 per km.

7 Common mistakes

- Comparing ratios when the two equations are not written in the same form.
- Sign errors when subtracting one equation from another. Change the sign of every term of the equation being subtracted.
- Stopping after finding one variable. A solution needs both x and y.
- Writing a two-digit number as xy instead of $10x + y$.

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